Sourabh Deshpande

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RESEARCH EXPERTISE

AI and Machine Learning for Digital Manufacturing and Industrial Automation

- Multiagent LLM-based digital twin for assisting factory workers
- Computer Vision for digitizing factory, predicting product quality and providing corrective feedback
- Human digital twin for safe human-machine interactions in the production and logistics facilities
- IIoT, Augmented / Virtual Reality for digital work instructions and machine control
- Synthetic 3D data generation for deep learning

EDUCATION

Ph.D., University of Cincinnati, OH, USA. GPA 4.0 (Advisor: Prof. Sam Anand) 2018 – Present
Dissertation: Machine Learning and IIoT-assisted Computer Vision Frameworks for Smart and Safe Manufacturing

M.S., Mechanical Engineering, Michigan Technological University, MI, USA. 2015 – 2017

B.S., Mechanical Engineering, University of Pune, India. 2010 – 2014

RESEARCH PROJECTS

Human Digital Twin Modeling for Improving Worker Safety in Manufacturing and Logistics — Jun '24 – Present

Ohio BWC (Bureau of Workers' Compensation) State Agency Research Contract
Partners: **Siemens Technology, Innovative Numerics, Worthington Steel, thyssenkrupp Bilstein**
Impact – Reducing non-fatal worker injuries (~75% as per NIOSH) by providing real-time ergonomic feedback to workers on the factory floor using cameras and smartwatch

- Leading a team of four researchers to develop a motion language model for assessing worker performance
- Trained an in-domain multi-task GPT model for automatically:
  - generating a 3D motion illustrating how to perform a new task ergonomically,
  - providing textual feedback on whether the worker's current motion could lead to injury,
  - predicting the worker's projected path to prevent any object collisions
- Devised LLM-assisted annotation toolbox to label custom human motions of factory workers with ergonomic descriptions
- Assisted Co-PIs Prof. Anand and Prof. Kumar in proposal writing of down-selected grant ($1.3M)

Enhancing Digital Twins and IoT for Legacy and Smart Factory Machines with Large Language Model (LLM) — Jan '24 – Dec '24

University of Cincinnati Industry 4.0/5.0 Institute Consortium Research Project
Partners: **Siemens Digital Industries, P&G, Kinetic Vision, Worthington Steel**
Impact – LLM-based Digital Twin for guiding new users to control factory machines (on-site/remotely) through voice commands; model expected to be deployed as a pilot project at Worthington Steel

- Co-developed a RAG-based multimodal multi-agent LLM framework to process IoT data for visualizing plots, generating documentation for machine operations, conducting fault diagnosis, and suggesting potential fixes (Latency < 1.5 seconds)
- Created an edge and cloud-based framework to display and control real-time legacy machine data on an iPad

Automated Inspection and Defect Quantification using AI and Computer Vision (CV) at Boeing — Jun '21 – Dec '22

MxD/DMDII (US Dept. of Army) Federal Research Contract (Lead UC student researcher for $1.6M grant; UC PI – Prof. Anand**)**
Partners: **Siemens Technology and Boeing**
Impact – Real-time process inspection with corrective feedback leading to an estimated savings of 560 human-hours per week

- Led UC research team to develop a closed-loop CV system for inspecting sealant application in space-grade solar panel
- Performed object localization and image segmentation of the sealant deposited on plates in real-time
- Trained a Bayesian network for causal inference for providing corrective action recommendations

Real-Time Visualization of Machine Sensors for Factory Maintenance using Cloud and AR Application — Jan '21 – Jun '22

Partner: **Volvo Trucks**
Impact – IIoT for real-time machine analytics on an AR app, reducing machine troubleshooting and downtime by 30%, translating to several million dollars per year in savings

- Co-developed an AR app (hosted on iPad) with sensor positions (spatial anchors) to visualize machine sensor values

Real-time component recognition using deep learning for worker training — Jan '23 – Dec '23

University of Cincinnati Industry 4.0/5.0 Institute Consortium Research Project
Partners: **Siemens Digital Industries, P&G, Kinetic Vision, Worthington Steel**
Impact – AR framework to simplify complex component identification, improving workforce training and efficiency

- Built a novel synthetic data generation pipeline based on CAD models using physics-based rendering methods
- Fine-tuned deep learning model (U-Net) and deployed on AR app to identify machine components and enable assembly

CAD-Based Computational Tools for Metal Powder Additive Manufacturing (AM) Jun '20 – Dec '20
Research contract with **Raytheon Technologies**
Impact – Identify and segment AM-specific features in CAD for powder-based fusion AM
- Developed computational geometry algorithms for DfAM feature recognition in 3D geometries; Designed frontend GUI within NX
- Assisted PI Prof. Anand in proposal write-up for $400K research contract

Prediction of 3D-printed Metal Part Quality using Hybrid Bayesian Network Jun '19 – Dec '20
Impact – Predict part quality based on AM process parameters to ensure parts meet quality standards 'right the first time'
- Built a Bayesian network to model the relationship between process variables and part quality in Selective Laser Melting (SLM)

Tata Motors Ltd. Dec '14 – May '15
Engineering Intern, Research and Development
- Designed rapid prototype and sheet metal components in CAD for 3D printing and press bending operations

PROGRAMMING AND SOFTWARE SKILLS

Programming languages: Python, C#, C++, MATLAB, JavaScript, HTML, SQLite
Commercial software: NX, Siemens Jack/Process Simulate, Ansys Additive, Autodesk Netfabb, Cortex
Open-source libraries: OpenCV, PyTorch, Tensorflow, Blender, ONNX, GitHub, HuggingFace

SELECTIVE PUBLICATIONS

IIoT-Enabled Digital Twin or Legacy and Smart Factory Machines with LLM Integration 2025
Anuj Gautam, Manish Aryal, **Sourabh Deshpande**, Shailesh Padalkar, Ming Tang, Sam Anand, et al.
SME North American Manufacturing Research Conference (Accepted)

Deep Learning-Based Recognition of Manufacturing Components using AR for Worker Training of Assembly Tasks 2024
Sourabh Deshpande, Manish Aryal, Sam Anand
ASME Manufacturing Science and Engineering Conference Proceedings, https://doi.org/10.1115/MSEC2024-125279

ImVR: Immersive Design Exploration and Process Integration for Additive Manufacturing of Complex Organic Geometries 2024
Manish Aryal, **Sourabh Deshpande**, Jillian Aurisano, Sam Anand
ASME Manufacturing Science and Engineering Conference Proceedings, https://doi.org/10.1115/MSEC2024-121253

Smart Monitoring and Automated Real-Time Visual Inspection of a Sealant Applications 2023
Sourabh Deshpande, Aditi Roy, Joshua Johnson, Ethan Fitz, Manish Kumar, Sam Anand
SME Manufacturing Letters, https://doi.org/10.1016/j.mfglet.2023.08.115

IIoT Based Framework for Data Communication And Prediction Using Augmented Reality For Legacy Machine Artifacts 2023
Sourabh Deshpande, Shailesh Padalkar, Sam Anand
SME Manufacturing Letters, https://doi.org/10.1016/j.mfglet.2023.08.058

IIoT Based Augmented Reality for Factory Data Collection and Visualization 2021
Jonathan Rosales, **Sourabh Deshpande**, Sam Anand
Procedia Manufacturing, https://doi.org/10.1016/j.promfg.2021.06.062

Prediction of Selective Laser Melting Part Quality Using Hybrid Bayesian Network 2020
Nathan Hertlein, **Sourabh Deshpande**, Vysakh Venugopal, Manish Kumar, Sam Anand
Additive Manufacturing, https://doi.org/10.1016/j.addma.2020.101089

RELEVANT COURSEWORK

- Intelligent Systems (AI), Introduction to Industrial AI, Statistical Methods
- Math Methods for Decision Engineering, Advanced Quality Engineering, Computational Methods in Additive Manufacturing

ACADEMIC MENTORSHIP

- Manish Aryal (M.S. Thesis) – ImVR: Enabling Immersive Exploration Environment for Complex Organic Geometries, 2023
- Anuj Gautam (M.S. Thesis) – IIoT-enabled Digital Twin with LLM Feedback for Manufacturing Applications, 2024

ACCOMPLISHMENTS AND LEADERSHIP EXPERIENCE

- University of Cincinnati Office of Research Digital Futures Fellowship, 2024
- Invention Disclosure (UC), along with PI Prof. Anand and Co-PI Prof. Kumar - Object Localization MxD, 2022
- Co-organized NAMRC and MSEC 2021 conference and recognized by SME and ASME; Received NSF student travel award
- Graduate teaching assistant, MECH 6071, Advanced Design for Manufacturing, 2019